



**Shri Vile Parle Kelavani Mandal's
USHA PRAVIN GANDHI COLLEGE OF ARTS,
SCIENCE AND COMMERCE(Autonomous)**



AI-RESCUE: INTELLIGENT DISASTER RESPONSE & RESOURCE COORDINATION PLATFORM

**A Field Project Report
Submitted in Partial Fulfillment
For Assessment in
Third Year
BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)**

Submitted by

Name: Dev Vikam

SAP ID: 53013240013

**Under the esteemed guidance of
Dr. Manisha Divate**

CERTIFICATE

This is to certify that Mr. Dev Vikam, SAP ID 53013240013, has successfully completed the Field Project Work in B.Sc. Information Technology under the Department of Information Technology with the project title "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" under the supervision and guidance of Dr. Manisha Divate.

The project focuses on the design and development of an intelligent, multi-agency emergency response and resource coordination platform that enables real-time incident reporting, citizen evidence ingestion, multi-modal AI triage analysis, severity and priority estimation, dynamic resource assignment (personnel, emergency vehicles, specialized equipment), healthcare facility bed tracking, evacuation shelter occupancy management, and automated alert dispatch.

We further certify that the work has been carried out by the student under proper academic guidance and that the project report has been prepared for academic assessment as per the requirements of the B.Sc. Information Technology curriculum.

Name and Signature of Guiding Teacher

Dr. Manisha Divate

Date of Submission: _____

COMPANY CERTIFICATE

This is to certify that Mr. Dev Vikam, a student of Third Year B.Sc. Information Technology (SAP ID: 53013240013) from Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), has successfully undergone and completed his Field Project work on the project entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" for a total duration of 90 Hours (comprising 30 Hours of Guide Supervision and 60 Hours of Core Engineering Work).

During the course of his field project, he exhibited high levels of dedication, technical proficiency, sound analytical acumen, and an ethical approach toward system design, database modeling, and software engineering. His contribution toward architectural specification, data modeling, and project documentation has been highly satisfactory.

Authorized Signatory & Organization Seal

Name & Designation: _____

Organization: Disaster Response & Crisis Tech Labs

Date: _____

DECLARATION

I, Mr. Dev Vikam, hereby declare that the project report entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" represents my original work carried out under the guidance of Dr. Manisha Divate.

The work presented in this report has been completed as part of the academic requirements of the Bachelor of Science in Information Technology program. This project has not been submitted previously to any other university or institution for the award of any degree, diploma, or certificate.

Wherever reference has been made to previous work, technologies, or resources, appropriate acknowledgement has been given. I further declare that the information presented in this report is true to the best of my knowledge and has been prepared in accordance with academic standards and ethical practices.

Name and Signature of the Learner

Dev Vikam

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LOG BOOK

Field Project – 90 Hours

Project Title: AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform

GUIDE SESSIONS – 30 HOURS

Date	Hours	Activity
12/06/2026	3	Project orientation and AI-Rescue overview; understanding the problem of fragmented disaster response and delayed resource allocation.
18/06/2026	3	Discussion of AI-Rescue objectives, scope, multi-agency applicability, user roles (Citizen, Response Officer, Admin), and expected outcomes.
20/06/2026	3	Survey of underlying architectural technologies: Leaflet/OSM GIS mapping, Spring Boot REST services, PostgreSQL database, and AI triage.
25/06/2026	3	Study of React web dashboard, WebSocket real-time broadcast services, relational schema normalization, and LLM/NLP incident triage integration.
27/06/2026	3	Requirement gathering: functional and non-functional requirements, citizen incident verification workflows, and officer triage dispatch rules.
02/07/2026	3	Review of emergency response criteria: severity scoring formulas, priority levels, hospital bed tracking, and evacuation shelter occupancy logistics.
23/07/2026	3	Planning and scheduling: Work Breakdown Structure (WBS), execution milestones, phased development lifecycle, and Gantt chart review.
13/08/2026	3	Review of conceptual models: architectural blueprints, Master Entity Relationship Diagram (ERD), DFD Level 0/1, and UML Use Case modeling.
29/08/2026	3	Review of database schema constraints, continuous system flowchart, UI wireframes, resource dispatch logic, and security audit log specifications.
24/09/2026	3	Final review of Chapter 1–4 documentation, architectural diagrams, test case scenarios, formatting consistency, and presentation preparation.

Total: 30 hours guide-supervised project work

WORK – 60 HOURS (Part 1: Initial Implementation)

Date	Hours	Activity
13/06/2026	3	Disaster management domain analysis, crisis response literature survey, and academic problem formulation.
16/06/2026	3	Comparative technology assessment of emergency frameworks, real-time alert dispatch, and GIS platforms.
19/06/2026	3	Formulation of AI-Rescue project charter, core objectives, and stakeholder requirement boundaries.
22/06/2026	3	Drafting Software Requirements Specification (SRS) for citizen, responder, officer, and admin modules.
26/06/2026	3	Citizen disaster reporting module UI wireframing and emergency evidence attachment workflow design.
29/06/2026	3	Response Officer dashboard layout, triage queue prioritization, and verification console planning.
04/07/2026	3	PostgreSQL relational schema modeling, 16 entity tables definition, and foreign key constraint specification.
08/07/2026	3	Master Entity Relationship Diagram (ERD) conceptual, logical, and physical cardinality modeling.
14/07/2026	3	Data Flow Diagram (DFD Level 0 Context & Level 1 Modular Decomposition) architectural modeling.
18/07/2026	3	UML Use Case Diagram modeling across 4 primary actors (Citizen, Officer, Admin, Responder) & 21 use cases.
25/07/2026	3	UML Sequence Diagram lifeline design modeling chronological procedural message passing across microservices.

WORK – 60 HOURS (Part 2: Design, Logic & Verification)

Date	Hours	Activity
31/07/2026	3	Continuous System Flowchart design modeling 41 operational nodes, decision diamonds, and error-retry bus loops.
06/08/2026	3	Multi-modal AI severity scoring algorithm formulation, confidence weighting, and priority calculation logic.
11/08/2026	3	Resource allocation engine, emergency vehicle capacity assignment, and equipment dispatch workflow.
18/08/2026	3	Healthcare facility bed availability index and evacuation shelter occupancy tracking service design.
24/08/2026	3	Real-time notification engine, emergency broadcast alerts, and role-based access control (RBAC) modeling.
02/09/2026	3	Security audit logging, system setting management, and data integrity constraint implementation.
08/09/2026	3	Test case scenario formulation covering authentication, incident validation, and AI dispatch workflows.
15/09/2026	3	Chapter 1 to Chapter 4 academic documentation drafting, table formatting, and cross-reference verification.
22/09/2026	3	Final report proofreading, high-resolution diagram vectorization, logbook synchronization, and submission binding.

Total Student Work: 60 Hours | Combined Field Project Total: 90 Hours Completed

Name & Signature of Company / Organization Name & Signature of Teacher

Disaster Response & Crisis Tech Labs

Dr. Manisha Divate

Name & Signature of Student

Dev Vikam